

Understanding How to Expose Clinical-Friendly Drug Cues with Virtual Reality and the Impact on Biofeedback Signals

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ABSTRACT

Cue exposure can be an invaluable tool by which to learn the skills of cue identification and craving perception in the development of mind-body awareness. Virtual reality (VR) has proven highly effective in cue-exposure therapy. In the current study, we developed a system for the collection of biofeedback data during the VR experience and while recounting these experiences with a psychotherapist afterward. Our results provide insight into the means by which VR cues induce cravings and impact the emotional state of the user.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Cue exposure; Drug addiction; Biofeedback.

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1 INTRODUCTION

Drug addiction is a complex issue that extends far beyond the compulsive use of drugs to include one's emotional state and the effects of drug use on relationships with friends, family members, and colleagues. Drug addiction can be treated from using psychiatric interventions as well as psychological interventions. Cognitive behavioral therapy (CBT) [3] emphasizes the identification and modification irrational thoughts, management of negative mood swings, and interventions to deal with lapses and other setbacks. Note however that most drug addicts either lack the capacity to identify the triggers eliciting drug cravings [3] or the motivation to deal with triggers once they are identified (impulse control issues). An inability to deal effectively with triggers often leads to behavioral responses aimed at satisfying the craving [9]. This cycle of triggers and responses can only be interrupted if the patient is aware of the cues that induce the cravings and corresponding bodily reactions. Cue exposure therapy (CET) mediates reactivity to behavioral cues by exposing the addict to drug-related stimuli with the aim of retraining their conditioned response to cues. Nonetheless, previous research has indicated that CET should be implemented in conjunction with other methods aimed at reducing the underlying desire for the stimulant.

Many technological innovations focus on the relationship between cue exposure and drug cravings [1]. VR has been proven particularly effective in cue-exposure therapy [2, 6], where it is used to generate virtual cues as an alternative to in vivo triggers. VR technology provides vivid visual reconstructions of cues within an immersive imaginary world featuring situations that would be difficult to reproduce in reality. Note however that the fidelity of these experiences can trigger powerful cravings, which the patient may find difficult to resist or disregard afterward [8].

This study explored the effects exposing patients to VR cues within the context of psychotherapy. We developed an in-lab system for collection of biofeedback signals during the VR experience. In this system, We created VR scenarios for sixteen types of exteroceptive and interoceptive cue identified in previous studies [5]. While the patient is watching the VR simulations, biofeedback signals are collected using wearable sensors. Each video session will end with the participants self-reporting on the strength of the cravings induced by the VR experience. The self-reports will then be compared with the corresponding biofeedback signals as an objective indicator of emotional responses.

2 IN-LAB DATA COLLECTION STUDY

2.1 Device

An HTC Cosmos Elite VR system will be used to display 3D virtual scenes depicting typical drug-related cues. At the same time, eVu TPS sensors [7] will be used to collect physiological signals, including heart rate variability (HRV), skin conductance (SC), and surface temperature (TEMP). The device is attached to a single finger using a fabric strap and the data is transmitted to a data logging application in real-time via Bluetooth.

2.2 Procedure

Figure 1 illustrates the progress of the VR/biofeedback experiment. One of the researchers will first explain to the participant the goal of the in-lab study, provide explanations of what to do during the VR experience, and then help them to put on the VR headset and biofeedback sensor. The transmission of biofeedback signals will be checked before initiating the VR system. The VR sequence will begin with the patient becoming familiar with the environment (60 sec). The patient will then be presented sixteen drug-related cues (150 sec each) with a brief question and answer session during the intervals (30 sec). The patient will then gradually exit the VR environment (60 sec).

2.2.1 Familiarization phase (one minute). It is important that participants transition smoothly from the physical to a virtual world to avoid disorientation and feelings of nausea. The participant will first be presented virtual representation of the lab space that they have just left. The participant will then be presented a VR tutorial on how to interact with the VR system, including the use of controllers. Once the participants gained familiarity with the system, they will be given instructions in how to use the controller to answer questions. Finally, the participant will click on a button to begin the CE experience. Note that the sixteen sequences will be presented in random order.

2.2.2 Exposure phase (two minutes for each cue). Our objective in this study will be to characterize the intensity of cravings elicited by the drug-related cues in VR. Each cue exposure will be implemented in the following three stages: (1) baseline recovery, (2) cue presentation, and (3) question-and-answer session.

Baseline recovery (one minute). It is important that the HRV, SC, and TEMP biofeedback signals begin from a standard baseline level to serve as a reference for the identification of changes in biofeedback readings in subsequent analysis. Thus, the VR experience will

begin with a VR tutorial on how to relax physically in order to return to a calm baseline state. Note that during the baseline recovery sequence, the participant will remain still in a standing position for 60 sec, during which baseline physiological signals will be collected.

Cue presentation (one minute). From previous studies [5], we have selected sixteen common drug-related cues. Exteroceptive cues include objects that resemble drug paraphernalia, lighters, wage receipts, messages from Social Networking apps, messages from drug dealers, night club music, messy hotel rooms, and friends hanging out in an environment typical of drug use. Interoceptive cues include scenarios that induce desire to enhance sex or feelings of low self-esteem, irritability, boredom, sadness, etc. Participants will be presented images of bad weather, immersed in browsing online profiles on dating apps, exposed to criticism, and subjected to work/school pressure. They will also be presented with situations that conjure up a desire to enhance sex or reward themselves via immediate gratification.

Question-answer sequences (30 seconds). During the Q&A sequences, the participants will be presented probing questions related to the cravings and emotions elicited by the previous VR cue. The answers will be presented in accordance with a visual analogue scale (VAS) [4] or seven-level Likert scale.

2.2.3 Exit phase (one minute). It is important that participants transition smoothly from the virtual world to the physical world. This will be achieved by reversing the familiarization phase, wherein the last VR image presented to the participant will be a virtual model of lab in which they are sitting. Participants will be encouraged to recount feelings of nausea from the VR experience or arousal from the cues. The participants will then be guided through a brief session of exercises aimed at emotional self-regulation.

2.3 Planned Measures & Data Analysis

Measures of craving strength will be derived from the self-reported recollections (Q&A) presented immediately after each cue presentation. Statistical characteristics (e.g., mean or variance) of those values and the biofeedback signals will be used to categorize the intensity of cravings, emotions, and feelings elicited by the respective cues.

3 FUTURE WORK & CONCLUSION

We are currently working with psychiatrists and psychologists to refine the design of the cue scenarios and post-cue questions. We will recruit methamphetamine-addicted patients via referrals from the psychiatrists with whom we are collaborating. The results collected in this pilot study are expected to shed light on the use of VR cues in psychotherapy.

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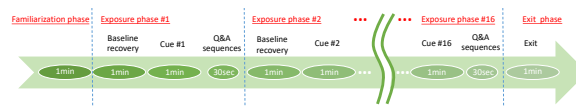


Figure 1: Flow chart showing an experiment involving multiple cue exposures

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